

Fateme GHAYEM

2025–Now	Postdoctoral research fellow TANGRAM team, Inria, Nancy, France
CONTACT INFORMATION	Email: fatemeh.ghayem@inria.fr fateme.ghayem@gmail.com Homepage: https://ghayem.github.io ORCID: https://orcid.org/0000-0001-5779-6685
EDUCATION	Université Grenoble Alpes , Grenoble, France Ph.D. in Signal, Image, Parole, and Télécoms, GIPSA-lab, Oct 2017 – Nov 2020 - Thesis topic: <i>Optimal sensor placement for source extraction</i> - Advisor: Prof. Christian Jutten, Dr. Bertrand Rivet Sharif University of Technology , Tehran, Iran M.Sc., Electrical Engineering, Sept 2013 – Sept 2015 GPA: 17.06/20 - Thesis topic: <i>MR image reconstruction from highly partial Fourier samples</i> - Advisor: Prof. Farokh Marvasti Shiraz University , Shiraz, Iran B.Sc., Electrical Engineering, Sept 2009 – Sept 2013 GPA: 17.85/20 National Organization for Development of Exceptional Talents , Shiraz, Iran Diploma, Mathematics and Physics, Sept 2005 – Sept 2009 GPA: 19.60/20
RESEARCH INTERESTS	– Image/text multi-modal foundation models for brain study – Brain meta-analysis with large language models (LLM) – Independent component/vector analysis (ICA/IVA) for multi-subject fMRI study – Aneurysm detection with deep learning – Optimal sensor placement for source extraction – Machine learning and statistical signal processing in medical applications – Bayesian modeling, numerical optimization, and dictionary learning
RESEARCH EXPERIENCES	• Postdoctoral researcher (2025–Now), TANGRAM, Inria Nancy, France. - Advisor: Dr. Erwan Kerrien - Research topic: Deep learning methods for aneurysm detection. - Summary: The project focuses on developing deep learning approaches for reliable aneurysm detection from MRA (Magnetic Resonance Angiography) and related modalities. The objectives are to: * Improve accuracy and robustness of aneurysm detection using advanced neural architectures; * Improve weakly annotated datasets for model training, as well as apply data augmentation techniques and multi-modal data analysis to address the data size limitation.

- **Postdoctoral researcher** (March 2023–2025), MIND, Inria Paris-Saclay, France.
 - Advisor: [Dr. Bertrand Thirion](#)
 - Co-advisor: [Dr. Demian Wassermann](#)
 - Research topic: Contrastive learning for neuroscience meta-analysis with rich text representation using LLMs
 - Summary:

The project focuses on large-scale brain meta-analysis, where the goal is to integrate neuroscientific literature, reported activation coordinates, and brain images in a unified predictive framework. We developed NeuroConText, a contrastive text-to-brain framework that leverages large language models to encode full-text articles and jointly learns text and image embeddings in a shared latent space. NeuroConText supports both retrieval, by finding studies and brain maps relevant to a free-text query, and encoding, by reconstructing activation maps from text, and has been shown to outperform established baselines such as NeuroQuery and Text2Brain. The main ideas in this work are to use modern representation learning to capture rich semantics and to address reproducibility and lack of statistical power in single neuroscientific studies by building scalable, data-driven meta-analytic tools. Furthermore, we introduced Peaks2Image, a neural approach that reconstructs whole-brain activation maps from sparse coordinates reported in articles, thereby bridging the gap between coordinate-based reports and image-based resources. Concretely, we:

 - * Learned a regression from neuroscience text to whole-brain activation maps, and a shared latent representation that supports both retrieval of relevant studies from text queries and reconstruction of brain maps from rich textual descriptions;
 - * Addressed key limitations of current automated meta-analysis tools, such as inconsistent terminology, limited handling of long texts and semantics, and sparse coordinate reporting by using large language models (LLMs), contrastive learning, and LLM-based text augmentation strategies;
 - * Supported second-level statistical synthesis by providing activation estimates associated with top- K retrieved studies as input to coordinate-based meta-analysis (CBMA), enabling more comprehensive and reproducible inferences about the relationship between brain function and behavior.
- **Postdoctoral researcher** (August 2021–August 2022), [MLSP-Lab](#), University of Maryland, Baltimore County (UMBC), Maryland, USA.
 - Advisor: [Prof. Tulay Adali](#)
 - Research topics:
 - Dictionary learning for the identification of new interpretable patterns and discriminative features from brain functional network connectivity (FNC) obtained from ICA decomposition of multi-subject resting-state fMRI data for static and dynamic studies.
 - Brain graph neural networks (Brain-GNN) for the classification of healthy control and patients with different brain disorders, *e.g.*, Schizophrenia.
 - Constrained ICA and IVA for subgroup identification from multisubject fMRI Data.
 - Reproducibility and replicability in neuroimaging data analysis.
- **Research assistant** (2015–2017), DSP-lab, EE Department, Sharif University of Technology, Tehran, Iran.
 - Advisor: [Prof. Massoud Babaie-Zadeh](#)
 - Research topics: Dictionary learning for sparse representation, convex/non-convex optimization.

- POSTERS “NeuroConText: Contrastive Text-to-Brain Mapping for Neuroscientific Literature”,
Organization for Human Brain Mapping (OHBM), Brisbane, Australia, 2025.
- WORKSHOP COGBASES workshop on open science methods for analyzing brain imaging data,
Paris, France, October 2023.
- SUMMER SCHOOL PRAIRIE artificial intelligence summer school (PAISS), Grenoble, France, July 2018.
- TALKS
- **From Contrastive Text-to-Brain Mapping to Multiview Learning of Depressive Disorders through Biological and Discourse Biomarkers**, MULTISPEECH, Inria-Nancy, France, February 2026.
 - **Multiview Learning of the Brain: Functional Connectivity, Subgroup Discovery, and Text-to-Brain Mapping**, IMoPA, ERCN, CHRU de Nancy, Nancy, France, February 2026.
 - **NeuroConText: Contrastive Learning for Neuroscience Meta-Analysis with Rich Text Representation**, *The British Council in Paris*, Paris, France, September 2025.
 - **Exploring Brain Function and Structure: From Sparse Coding to Multimodal Meta-Analysis**, *Laboratoire de Physique de l'ENS de Lyon*, Lyon, France, February 2024.
 - **Multi-subject fMRI Analysis for Brain Pattern Discovery and Subgroup Identification**, *MIND team, Inria-Saclay*, Paris, France, June 2024.
 - **New Interpretable Patterns and Discriminative Features from Brain Functional Network Connectivity using Dictionary Learning**, *MIND team, Inria-Saclay*, Paris, France, September 2023.
 - **Optimal Sensor Placement for Source Extraction**, *MIND team, Inria-Saclay*, Paris, France, June 2023.
 - **Optimal Sensor Placement for Source Extraction**, *Diagnostic and Interventional Adaptive Imaging (IADI)*, Nancy, France, January 2023.
 - **Optimal Sensor Placement for Source Extraction**, *Centre de Recherche en Automatique de Nancy (CRAN), Department of Biology, Signals and Systems*, Nancy, France, January 2021.

Publications

- PREPRINTS 1. R. Meudéc, J. Dockès, **F. Ghayem**, D. Wassermann, B. Thirion, “Peaks2Image: Reconstructing fMRI Maps from Stereotactic Coordinates to Enhance Cognitive Meta-Analysis,” submitted to *Imaging Neuroscience*, August 2025. [\[Paper\]](#)
- REFEREED JOURNAL PUBLICATIONS 1. **F. Ghayem**, R. Meudéc, J. Dockès, B. Thirion, D. Wassermann, “NeuroConText: Contrastive Learning for Neuroscience Meta-Analysis with Rich Text Representation,” *Imaging Neuroscience*, 2026.
2. **F. Ghayem**, B. Rivet, C. Jutten, R. Cabral Farias, “Robust sensor placement for signal extraction”, *IEEE Transactions on Signal Processing*, vol. 69, pp. 4513-4528, 2021.

CONFERENCE
PUBLICATIONS

3. **F. Ghayem**, M. Sadeghi, M. Babaie-Zadeh, S. Chatterjee, M. Skoglund, and C. Jutten, "Sparse signal recovery using iterative proximal projection", *IEEE Transactions on Signal Processing*, vol. 66, no. 4, pp. 879–894, February 2018.
1. R. Meudec*, **F. Ghayem***, J. Dockès, D. Wassermann, B. Thirion, "NeuroConText: Contrastive Text-to-Brain Mapping for Neuroscientific Literature", *International Conference on Medical Image Computing And Computer Assisted Intervention (MICCAI)*, October 2024.
* **Equal contribution**
2. **F. Ghayem**, H. Yang, F. Kantar, S-J. Kim, V. D. Calhoun, T. Adali, "New Interpretable Patterns and Discriminative Features from Brain Functional Network Connectivity Using Dictionary Learning", *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Rhodes island, June 2023.
3. H. Yang, **F. Ghayem**, B. Gabrielson, M. A. B. S. Akhonda, V. D. Calhoun, T. Adali, "Constrained independent component analysis based on entropy bound minimization for subgroup identification from multisubject fMRI data", *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, June 2023.
4. H. Yang, MABS. Akhonda, **F. Ghayem**, Q. Long, V.D. Calhoun, T Adali, "Independent Vector Analysis Based Subgroup Identification from Multisubject fMRI Data", in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2022.
5. **F. Ghayem**, B. Rivet, Ch. Jutten, R. Cabral Farias, "Gradient-based algorithm with spatial regularization for optimal sensor placement", in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2020.
6. **F. Ghayem**, B. Rivet, Ch. Jutten, R. Cabral Farias, "Optimal sensor placement for signal extraction", in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2019.
7. M. Sadeghi, **F. Ghayem**, M. Babaie-Zadeh, S. Chatterjee, M. Skoglund, and C. Jutten, "L0Soft: ℓ_0 Minimization via Soft Thresholding", in *Proceedings of the 27th European Signal Processing Conference (EUSIPCO)*, 2-6 September 2019.
8. **F. Ghayem**, M. Sadeghi, M. Babaie-Zadeh, and C. Jutten, "Accelerated dictionary learning for sparse signal representation", in *13th International Conference on Latent Variable Analysis and Signal Separation, LVA/ICA*, Grenoble, France, 2017.
9. **F. Ghayem** and F. Rassaie, "Helical antenna to measure radiated power density around a BTS; Design and implementation", in *third Asia-Pacific Conference on Antennas and Propagation (APCAP)*, July 2014.

CO-SUPERVISION

I am currently supervising four Master students at Université de Lorraine on a project titled "Enhanced NeuroConText with deep image representation for brain meta-analysis". Also, I co-supervised three Ph.D. students in collaboration with Prof. Tulay Adali (University of Maryland, Baltimore County, USA), Prof. Jean-Christophe Pesquet (CentraleSupélec, Université Paris-Saclay, France), Prof. Vince D. Calhoun (Translational Research in Neuroimaging and Data Science, USA), and Dr. Seung-Jun Kim (University of Maryland, Baltimore County, USA) on the following projects:

- Brain graph neural networks (Brain-GNN) for the classification of healthy control and patients with different brain disorders, e.g. Schizophrenia.
- ICA and IVA for subgroup identification from multisubject resting state fMRI Data.
- Reproducibility and replicability in neuroimaging data analysis.

TEACHING	Computer Vision and Image Processing	2026
	- 20 hours practical sessions	
	- Polytech Nancy	
	Signal Processing	2025-2026
	- 6 hours lectures + 12 hours practical sessions	
	- Université de Lorraine	
	Signal Processing	2024-2025
	- 6 hours lectures + 12 hours practical sessions	
	- Université de Lorraine	
	Signals and Systems	2015
HONORS & AWARDS	- 10 hours practical sessions	
	- Responsible: Prof. M. Babaei-Zadeh, Sharif University of Technology	
	Digital Signal Processing II	2014
	- 10 hours practical sessions	
	- Responsible: Prof. F. Marvasti, Sharif University of Technology	
	Signals and Systems	2013
	- 20 hours practical sessions	
	- Responsible: Dr. M. Derakhtian, Shiraz University	
	Electromagnetics	2012
	- 20 hours practical sessions	
COMPUTER SKILLS	- Responsible: Dr. M. Derakhtian, Shiraz University	
	Electrical Circuit II	2011
	- 20 hours practical sessions	
	- Responsible: Prof. M. A. Masnadi-Shirazi, Shiraz University	
COMMUNITY SERVICES		
QUALIFICATIONS	I am qualified to serve as Maître de Conférences, Section 26 – Mathématiques appliquées et applications des mathématiques.	
COMPUTER SKILLS	• <i>Programming Languages and Softwares:</i> Python, PyTorch, MATLAB	2024
	• <i>Typesetting:</i> $\text{\LaTeX}$	2017
	• <i>Full travel grant (CHESS project), LVA/ICA workshop, Grenoble, France.</i>	2017
	• <i>Bronze award in math competition among high school students,</i>	2008
	Sharif University of Technology.	
	• <i>Admitted to National Organization for Development of Exceptional Talents</i>	2005
	(NODET) as high school and pre-university school student.	
COMMUNITY SERVICES	Reviewer for the following journals and conferences:	
	- AAAI	2026
	- ICML	2025
	- NeurIPS	2024-2025
	- Journal of Machine Learning Research	2024-2025
	- IEEE Transactions on Machine Learning Research	2024-2025
	- IEEE Transactions on Medical Imaging	2023-2026
	- IEEE Transactions on Signal Processing	2019-2022
	- IEEE Signal Processing Letters	2019, 2021, 2023
	- International Conf. on Acoustics, Speech, and Signal Proc. (ICASSP)	2023
	- European Signal Processing Conference (EUSIPCO)	2019-2021

LANGUAGE	- English (Fluent)
PROFICIENCY	- French (Intermediate)
	- Persian (Native)

HOBBIES AND INTERESTS	Playing the violin, running, hiking, biking
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REFERENCES	• Prof. Christian Jutten christian.jutten@gipsa-lab.grenoble-inp.fr • Prof. Massoud Babaie-Zadeh mbzadeh@sharif.edu • Dr. Bertrand Thirion bertrand.thirion@inria.fr • Dr. Demian Wassermann demian.wassermann@inria.fr • Dr. Bertrand Rivet bertrand.rivet@gipsa-lab.grenoble-inp.fr • Dr. Erwan Kerrien erwan.kerrien@inria.fr • Prof. Tulay Adali Adali@umbc.edu
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